IS6055: Prescriptive Analytics Assignment

Assignment Type: Prescriptive Analytics (Group Project)
Percentage of Module Mark: 50%
Due Date: on or before Monday 7th April 2025

Assignment Title: Sustainable Supply Chain Optimization for GreenGlow Cosmetics

Abstract:

GreenGlow Cosmetics, a multinational company specializing in skincare and beauty products, is scaling operations to meet increasing demand while adhering to stringent environmental regulations. The company operates four production plants and distributes products to six global regions. Challenges include meeting fluctuating demand, minimizing costs, and reducing environmental impact under uncertain supply chain conditions.

To address these challenges, your team will develop a data-driven optimization model that considers several critical factors:

1. Supplier capabilities, including material costs, shipping modes, and delivery variances (Table 1).
2. Production capacities and operational efficiencies of GreenGlow's regional plants (Table 2).
3. Demand patterns and delivery penalties across six regions, reflecting customer satisfaction and fluctuating requirements (Table 3).
4. Transportation costs and emissions, highlighting inter- and intra-regional cost structures (Table 4).

Your goal is to optimize three competing objectives:

1. Minimize total costs (procurement, production, and transportation).
2. Reduce environmental impact (CO_2 emissions from transportation and energy usage).
3. Maximize customer satisfaction by meeting fluctuating regional demands with on-time delivery.

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Additional Considerations:

- Suppliers may deliver between 80%-120% of their monthly quotas.
- Demand in each region fluctuates within ±10% of forecasted levels.

Assignment Tasks:

Part 1: Problem Analysis

a. Analyze the supply chain, identifying key bottlenecks and risks.

b. Discuss potential trade-offs between cost, sustainability, and service quality.

Part 2: Optimization Model Development

a. Decision Variables
   - Raw material quantities sourced from each supplier.
   - Units of each product produced in each plant.
   - Quantities transported between plants, warehouses, and regions.
   - Whether to expand operations (binary decision).

b. Objective Function
   - Minimize costs, CO2 emissions, and unmet demand penalties using a weighted-sum multi-objective approach.

c. Constraints
   - Supply constraints: Supplier availability limits and delivery uncertainty.
   - Capacity constraints: Production limits of each plant and transportation network.
   - Demand satisfaction: Meet regional demand within specified fluctuations.

- Environmental constraints: CO2 emissions from transportation and energy usage must not exceed specified thresholds.

d. Advanced Optimization Techniques:

- Solve using Mixed-Integer Linear Programming (MILP).
- Incorporate stochastic programming to handle uncertainty.
- Utilize network flow and transportation modeling for transportation optimization.

Part 3: Scenario Analysis

a. Sensitivity Analysis:

- How changes in supplier costs, demand levels, or transportation costs impact the optimal solution.

Part 4: Recommendations and Visualization

a. Present Actionable Recommendations
   - Prioritize Key Insights: Highlight the most critical findings from the optimization model. For example:
     - Which suppliers should be prioritized based on cost, reliability, and environmental impact?
     - Which production plants should be scaled up or down to meet regional demand while minimizing costs and emissions?
     - How should transportation routes be structured to balance costs and CO2 emissions?
   - Propose Specific Actions: Provide clear recommendations, such as:
     - Increase procurement from Supplier X during peak demand seasons due to lower costs and higher reliability.
     - Shift 20% of production to the Asia plant to take advantage of lower operating costs.

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b. Discuss Trade-Offs and Constraints
   - Cost vs. Environmental Impact:
     - Analyze situations where minimizing costs may increase CO2 emissions and vice versa.
     - Suggest thresholds or policies (e.g., a maximum allowable emission level) to balance these trade-offs.
   - Capacity Limitations:
     - Discuss potential bottlenecks in plant production or transportation routes.
     - Recommend solutions such as expanding specific plants or investing in additional shipping capacity.

c. Visualize Results Effectively (<u>if possible</u>)
   - Graphs and Charts: Use visualizations to enhance understanding:
     - Heatmaps for transportation costs and CO2 emissions between regions.
     - Bar charts comparing production costs and emissions across plants.
     - Pie charts showing the proportion of demand met by each plant or supplier.
   - Geographic Maps: Use maps to illustrate:
     - Transportation routes and their associated costs.
     - Regional demand fulfillment and potential bottlenecks.

d. Provide Contingency Plans
   - Supplier Disruptions:
     - Explain how to handle cases where a supplier cannot deliver.
     - Suggest alternative sourcing strategies and their cost implications.
   - Plant Shutdowns:
     - Analyze how production can be redistributed if a plant goes offline temporarily.
     - Discuss the financial and logistical impacts.
   - Demand Surges:
     - Propose strategies to manage unexpected spikes in demand, such as:
       - Increasing buffer stock in high-demand regions.

- Leveraging additional shipping capacity for faster delivery.

e. Evaluate Sustainability Metrics
   - CO2 Reduction Goals:
     - Quantify the environmental benefits of the proposed solution (e.g., "CO2 emissions are reduced by 20% compared to the baseline scenario").
   - Energy Efficiency:
     - Highlight improvements in energy usage across production plants.
   - Alignment with Sustainability Objectives:
     - Discuss how the recommendations support GreenGlow's commitment to environmental responsibility.

f. Support Recommendations with **Evidence**
   - Statistical Summaries:
     - Include key metrics, such as cost savings, percentage of demand fulfilled, and total CO2 emissions.
   - Scenario Analysis Results:
     - Present findings from sensitivity analyses to demonstrate the robustness of the recommendations.
   - Literature or Industry Benchmarks:
     - Compare the results to industry standards or case studies from similar organizations.

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Supporting Data:

Table 1: Raw Material Suppliers

Supplier Name	Material Type	Cost per Unit ($)	Monthly Supply (kg)	Shipping Mode	CO2 Emissions (kg/unit)	Delivery Variance (%)
Nature's Harvest	Organic Oils	15	250,000	Sea/Air	0.5 (Sea)/1.0 (Air)	±20%
Pure Essence	Botanical Extracts	25	150,000	Air	1.5	±10%
Mineral Glow	Mineral Bases	10	400,000	Sea	0.8	±15%
Floral Fusion	Floral Scents	20	200,000	Air/Sea	1.0 (Air)/0.6 (Sea)	±10%

Note: This table summarizes the suppliers of GreenGlow Cosmetics, detailing the material type, cost per unit, monthly supply capacity, available shipping modes, associated CO2 emissions for each shipping option, and delivery variance percentages, highlighting supply chain constraints and environmental impact factors.

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Table 2: Production Plants

Region	Monthly Capacity (Units)	Operating Cost per Unit ($)	Energy Usage (kWh/unit)	CO2 Emissions (kg/unit)
Europe	800,000	12	2.5	0.8
North America	1,200,000	10	2.2	0.7
Asia	1,500,000	8	2.0	0.6
South America	600,000	9	2.3	0.9

Note: this table outlines the production capacities of GreenGlow's plants across regions, specifying their monthly unit capacity, operating costs, energy usage per unit, and CO2 emissions, providing insights into cost-efficiency and environmental impact for each plant.

Table 3: Regional Demand (Monthly)

Region	Demand for Product A	Demand for Product B	Demand for Product C	Delivery Penalty ($/unit)	Demand Variance (%)
North America	500,000	400,000	300,000	5	±10%
Europe	300,000	450,000	350,000	6	±10%
Asia-Pacific	600,000	500,000	400,000	4	±15%

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South America	200,000	300,000	200,000	6	±10%
Middle East	250,000	400,000	300,000	7	±10%
Africa	150,000	200,000	150,000	8	±15%

Note: this table presents the regional demand for GreenGlow's products (A, B, and C), along with delivery penalty costs per unit for unmet demand and the percentage variance in demand, highlighting the fluctuations and costs associated with meeting customer needs in each region.

Table 4: Transportation Costs and Emissions (per Unit)

From/To	North America	Europe	Asia-Pacific	South America	Middle East	Africa
North America	($0.7 kg)	3	5	2	6	7
Europe	3	($0.6 kg)	4	5	2	3
Asia-Pacific	5	4	($0.5 kg)	6	7	8
South America	2	5	6	($0.8 kg)	7	9

Note: this table displays the transportation costs between regions for GreenGlow's supply chain, including per-unit costs in dollars for moving products, with specific values for inter-regional transport and associated costs per kilogram for intra-regional shipments.

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Note:

- **These tasks must be implemented using Python. You need to submit your Python code along with your report.**

Required Format:

a) 6,000 words excluding references and statistical summaries or programming codes.
b) All external sources of information must be referenced.
c) Font – Times New Roman size 12.
d) Spacing – 1.5.

- **Referencing Style:** For this course, you should use the **Harvard referencing style**. This includes both in-text citations and a reference list at the end of your assignment.
- **In-Text Citations:** When you refer to or paraphrase someone else's work within your text, you must include an in-text citation. Here are a couple of examples:
    - **Book**: (Ladley, 2020)
    - **Journal Article**: (Khatri & Brown, 2010)
    - **Website**: (Data Governance Institute, 2023)

- **Reference List:** At the end of your assignment, include a full list of references with all the details of each source. Examples:
    - **Book**: Ladley, J. (2020) *Data Governance: How to Design, Deploy, and Sustain an Effective Data Governance Program*. 2nd ed. Waltham: Elsevier.
    - **Journal Article**: Khatri, V. and Brown, C.V., 2010. Designing data governance. *Communications of the ACM*, *53*(1), pp.148-152.
    - **Website**: Data Governance Institute (2023) *The Data Governance Framework*. Available at: https://www.datagovernance.com/framework (Accessed: 15 August 2024).

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Group Formation Instructions

- Group Size:
  Each group must consist of **6 or 7** members.
- Group Roles:
  Each group member should take responsibility for a specific aspect of the assignment.
- Schedule regular meetings to discuss progress and resolve challenges.
- Use collaborative tools like Microsoft Teams to work together efficiently.
- Create a timeline with milestones for each part of the assignment.
- If you are unable to find a group, please notify me by **the end of Week 3**. You will be assigned to a group based on availability.

Submission & Formatting Guidelines:

When submitting your assignment, please ensure that your file is properly formatted and named according to the following guidelines:

- **File Name Format:**
  - IS6055-Group Assignment-Your group number
    - Example: IS6055- Group Assignment-Group1

- **File Type:**
  - Please submit your work as a **PDF** or **Word document (.docx)** along with the Python file (PyCharm or Jupyter)
- **File Content:**
  - **Title Page:** Include a title page with the following information:
    - Module Code: IS6055
    - Module Title: Prescriptive Analytics
    - Group Number (e.g., IS6055-Group1)
    - Full Names and Student IDs of all members
    - Primary Contact for the group (email/phone)
    - Submission Date

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Footers: Include page numbers in the footer of each page.

- **Submit your assignment as a Word/PDF document along with the Python file (PyCharm or Jupyter) through Canvas.**
- **All assignments will be checked for plagiarism using Turnitin.**
- **All assignments will be graded according to the rubric provided in Table-Rubric.**

Table Rubric: Assessment Criteria for Sustainable Supply Chain Optimization Assignment

Criteria	Excellent (85-100%)	Good (70-84%)	Satisfactory (50-69%)	Needs Improvement (30-49%)	Unsatisfactory (0-29%)
1. Problem Analysis and Trade-Offs (15%)	Thorough analysis of supply chain challenges and trade-offs. All major issues are identified, and detailed explanations are provided. Clear understanding of cost, sustainability, and service quality trade-offs.	Good analysis with most challenges identified. Trade-offs are explained but lack some depth or clarity.	Basic analysis with limited identification of challenges. Trade-offs are mentioned but not well-explained.	Incomplete or unclear analysis of challenges. Trade-offs are poorly explained or missing key points.	No or very poor analysis of challenges. Trade-offs are absent or completely irrelevant.
2. Optimization Models (30%)	Correctly formulated models with advanced methods (e.g., MILP, stochastic programming). Constraints and assumptions are clear and logical. Implementation is robust, and results are well-supported by evidence.	Models are well-formulated with minor errors or omissions. Use of advanced methods is evident but may lack depth or clarity in some areas. Constraints and assumptions are mostly appropriate.	Models are basic with some errors or missing components. Advanced methods are used but not well-implemented or justified. Constraints and assumptions are partially addressed.	Models are poorly formulated, with significant errors or missing components. Limited use of advanced methods. Constraints and assumptions are unclear or poorly justified.	Models are absent, incorrect, or irrelevant. No use of advanced methods. Constraints and assumptions are missing or nonsensical.
3. Sensitivity Analysis (15%)	Multiple scenarios are explored thoroughly, and the robustness of	Scenarios are explored with reasonable depth,	Limited exploration of scenarios.	Minimal exploration of scenarios. Analysis of robustness is shallow or	Scenarios are not explored. No analysis of robustness is

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	the solution is analyzed with depth. Results are well-documented and explained with supporting evidence and visualizations.	but some areas lack detail. Robustness of the solution is analyzed but with minor gaps in explanation or evidence.	Robustness is discussed but not fully analyzed or supported by evidence. Visualizations may be basic or unclear.	poorly supported. Visualizations are missing or irrelevant.	conducted. Visualizations are absent or incorrect.
4. Recommendations and Visualizations (30%)	Recommendations are actionable, insightful, and supported by robust analysis. Visualizations are clear, professional, and enhance understanding. Sustainability metrics and contingency plans are comprehensive and practical.	Recommendations are clear and actionable, with minor gaps in analysis or evidence. Visualizations are effective but may lack some professional polish or detail. Sustainability metrics and contingency plans are addressed but lack depth.	Recommendations are basic and lack actionable insights. Visualizations are simple and provide limited value. Sustainability metrics and contingency plans are minimally addressed or unclear.	Recommendations are vague or impractical. Visualizations are poorly constructed or do not enhance understanding. Sustainability metrics and contingency plans are superficial or missing.	Recommendations are absent, irrelevant, or unsupported. Visualizations are missing, irrelevant, or incomprehensible. Sustainability metrics and contingency plans are not addressed.
5. Formatting and Referencing (5%)	All formatting guidelines are followed precisely. Referencing is thorough, consistent, and correctly formatted in the Harvard style.	Formatting guidelines are followed with minor errors. Referencing is mostly complete and correctly formatted, with a few inconsistencies.	Formatting guidelines are followed but with noticeable errors. Referencing is incomplete or contains formatting errors in several instances.	Formatting guidelines are poorly followed. Referencing is inadequate or contains multiple formatting errors.	Formatting guidelines are not followed. Referencing is missing or incorrect.
6. AI Supervisory Skills (5%)	Demonstrates exceptional ability to supervise AI tools effectively. Critically evaluates and refines AI-generated content. Clear evidence of supervision and decision-making in reports and code.	Shows good ability to supervise AI tools. Evaluates AI output but may miss minor issues. Evidence of supervision and decision-making is present but could be more detailed.	Demonstrates basic AI supervision skills. Occasionally accepts AI output without critical evaluation. Evidence of supervision is	Shows limited ability to supervise AI tools. Often accepts AI output without critical evaluation. Evidence of supervision is vague or insufficient.	Fails to supervise AI tools effectively. Blindly accepts AI output without evaluation. Evidence of supervision is absent or unconvincing.

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			minimal or unclear.		

AI Supervisory Skills:

Note: If you choose to use **AI tools like ChatGPT** to assist with your assignment, you must provide transparency in your work. Specifically, if you incorporate any content generated by ChatGPT or similar AI tools, you are required to share the link to the specific conversation or session with me along with your assignment submission. This ensures that all sources of information are properly acknowledged, and it helps maintain the integrity of your work. Failure to disclose the use of such tools may be considered a breach of academic honesty policies.

- As you interact with AI tools like ChatGPT, your task is not only to use the tool but to critically supervise the suggestions it provides. This means evaluating the accuracy, relevance, and applicability of AI-generated content. You should refine and adapt the AI's suggestions to better suit your assignment's context.
- You are required to share the code and the final report with clear documentation on how you used AI assistance. Your documentation should explain how you supervised the AI, what decisions you made to modify or accept its suggestions, and why.
- Your ability to supervise the AI will be evaluated based on how well you refine and adapt the AI-generated content. You will be assessed on your critical thinking, decision-making, and the quality of your final outputs. The emphasis will be on ensuring that you use AI as a tool to enhance your work, not as a crutch to produce uncritical or unrefined results.

Feedback Mechanisms

INTERIM FEEDBACK. Project groups may request feedback on progress throughout the course of the project via the following mechanisms: (1) Scheduled meeting with lecturer or (2) Email feedback on draft submissions.

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Penalties

Submission of a project indicates that this is the student's own work. If the submission is found not to be original work, a mark of 0% will be automatically awarded. This project is worth 50% of the total module percentage for IS6052. Where work is submitted up to and including 7 days late, 10% of the total marks available shall be deducted from the mark achieved. Where work is submitted up to and including 14 days late, 20% of the total marks available shall be deducted from the mark achieved. Work submitted 15 days late or more shall not be accepted.

This project outline adheres (as appropriate) to the policies and codes of conduct outlined in Handbook Governing Curriculum Approval, the Policies and Guidelines Governing Academic Programmes, DMIS Guidelines, the Guide to Examinations for Staff and Students, the Plagiarism Policy, the Duty of Respect and Right to Dignity Policy, and the UCC Student Rules etc.